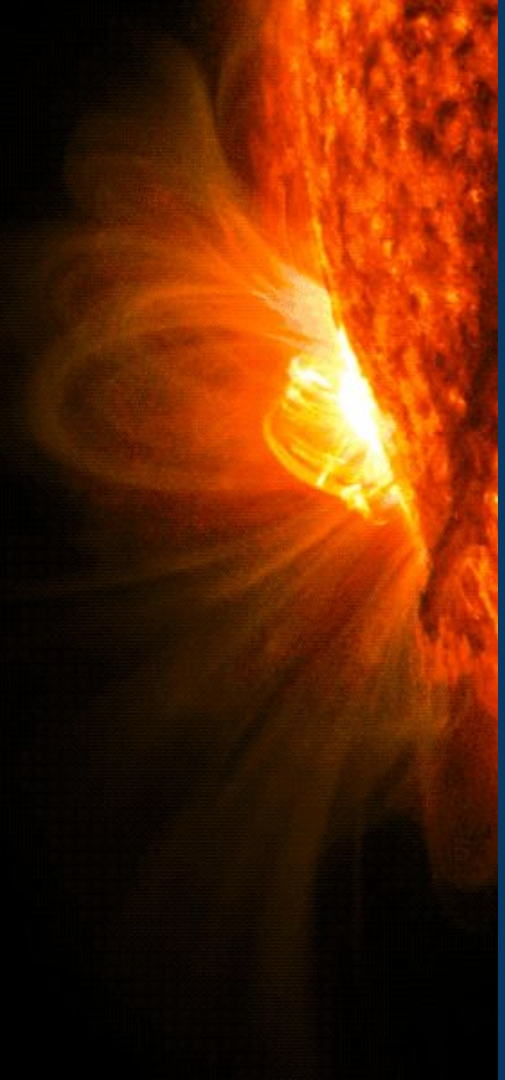
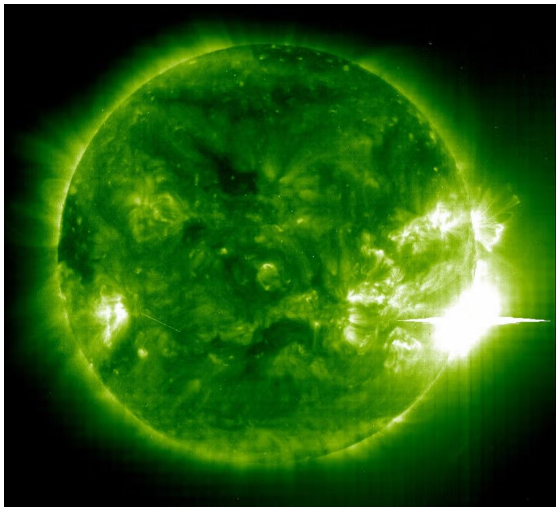


Modeling Electron Transport and Hard X-ray Polarization in Solar Flares

Milena Barrios-Gallego

Why HXR polarization matters?

Solar flares are sudden, intense explosions that release energy and radiation on the Sun's surface.



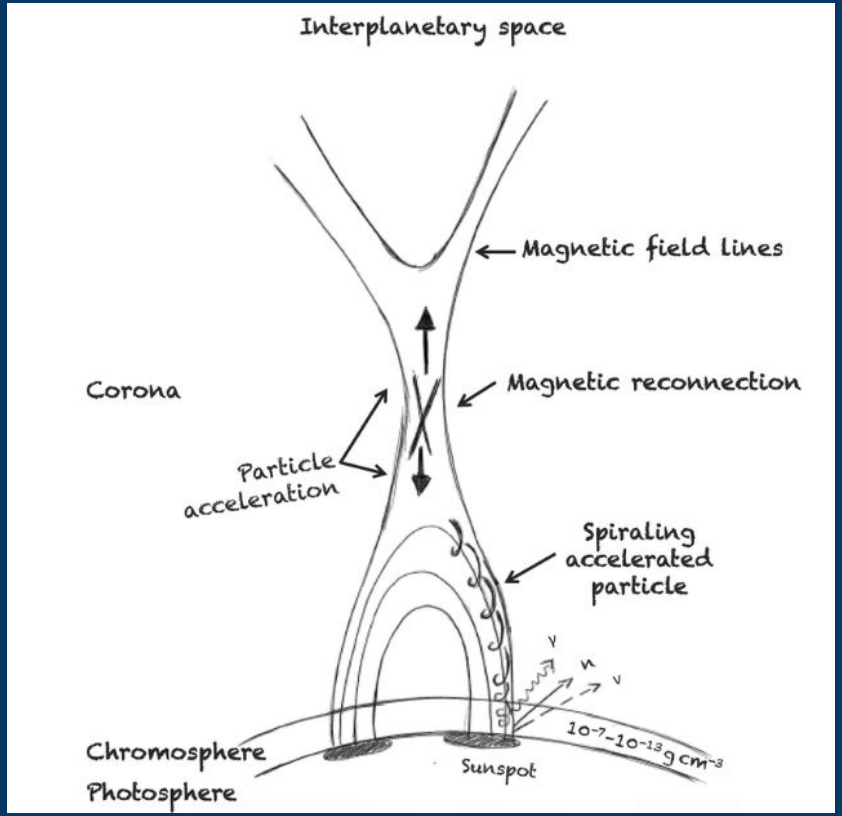
What we observe:

Hard X-ray emission… Bremsstrahlung!

The observed radiation contains information about the electron distribution

What we infer!

- Electron energy spectrum
- Location of energy deposition
- Electron directivity



Taken from: IceCube Collaboration. (2021). Search for GeV neutrino emission during intense gamma-ray solar flares with the IceCube Neutrino Observatory. Physical Review D, 103(10), 102001. [link removed]

Physical Setup

- Magnetic reconnection in the corona accelerates electrons to high energies.
- Accelerated electrons spiral along converging magnetic field lines toward the chromosphere.
- Magnetic mirroring can reflect part of the electron population back upward.
- Electrons interact with the ambient plasma and induce a return-current electric field.
- Electrons that reach the chromosphere emit hard X-rays.

From Electron Transport to the Fokker Planck Equation

The electron beam is described by a distribution function f=f(t,z,E,μ), where

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\rm coll} + \left(\frac{\partial f}{\partial t}\right)_{\rm magn},$$

z: depth along the magnetic loop

E: electron kinetic energy

α: pitch angle

μ = cosα

The goal is to evolve f under:

- Streaming
- Electric Field
- Collisions
- Magnetic Convergence

A&A 512, A8 (2010)
DOI: 10.1051/0004-6361/200811486
© ESO 2010

Astronomy & Astrophysics

Diagnostics of energetic electrons with anisotropic distributions in solar flares

I. Hard X-rays bremsstrahlung emission

V. V. Zharkova[1], A. A. Kuznetsov[2], and T. V. Siversky[1]

[1] Department of Computing and Mathematics, University of Bradford, Bradford BD7 1DP, UK
e-mail: [v.v.zharkova;t.siverskyi]@brad.ac.uk
[2] Institute of Solar-Terrestrial Physics, Irkutsk, Lermontov str. 126a, 664033, Russia
e-mail: a_kuzn@iszf.irk.ru

From Electron Transport to the Fokker Planck Equation

The electron beam is described by a distribution function f=f(t,z,E,μ), where

z: depth along the magnetic loop

E: electron kinetic energy

α: pitch angle

μ = cosα

Free streaming of the e^- along the magnetic field

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\rm coll} + \left(\frac{\partial f}{\partial t}\right)_{\rm magn},$$

A&A 512, A8 (2010)
DOI: 10.1051/0004-6361/200811486
© ESO 2010

Astronomy & Astrophysics

The goal is to evolve f under:

- Streaming
- Electric Field
- Collisions
- Magnetic Convergence

Diagnostics of energetic electrons with anisotropic distributions in solar flares

I. Hard X-rays bremsstrahlung emission

V. V. Zharkova[1], A. A. Kuznetsov[2], and T. V. Siversky[1]

[1] Department of Computing and Mathematics, University of Bradford, Bradford BD7 1DP, UK
e-mail: [v.v.zharkova;t.siverskyi]@brad.ac.uk
[2] Institute of Solar-Terrestrial Physics, Irkutsk, Lermontov str. 126a, 664033, Russia
e-mail: a_kuzn@iszf.irk.ru

From Electron Transport to the Fokker Planck Equation

The electron beam is described by a distribution function f=f(t,z,E,μ), where

z: depth along the magnetic loop

E: electron kinetic energy

α: pitch angle

μ = cosα

Ohmic losses: energy losses due to the self-induced electric field

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\rm coll} + \left(\frac{\partial f}{\partial t}\right)_{\rm magn},$$

The goal is to evolve f under:

- Streaming
- Electric Field
- Collisions
- Magnetic Convergence

A&A 512, A8 (2010)
DOI: 10.1051/0004-6361/200811486
© ESO 2010

Astronomy & Astrophysics

Diagnostics of energetic electrons with anisotropic distributions in solar flares

I. Hard X-rays bremsstrahlung emission

V. V. Zharkova[1], A. A. Kuznetsov[2], and T. V. Siversky[1]

[1] Department of Computing and Mathematics, University of Bradford, Bradford BD7 1DP, UK
e-mail: [v.v.zharkova;t.siverskyi]@brad.ac.uk
[2] Institute of Solar-Terrestrial Physics, Irkutsk, Lermontov str. 126a, 664033, Russia
e-mail: a_kuzn@iszf.irk.ru

From Electron Transport to the Fokker Planck Equation

The electron beam is described by a distribution function f=f(t,z,E,μ), where

z: depth along the magnetic loop

E: electron kinetic energy

α: pitch angle

μ = cosα

Pitch angle change

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\rm coll} + \left(\frac{\partial f}{\partial t}\right)_{\rm magn},$$

A&A 512, A8 (2010)
DOI: 10.1051/0004-6361/200811486
© ESO 2010

Astronomy & Astrophysics

The goal is to evolve f under:

- Streaming
- Electric Field
- Collisions
- Magnetic Convergence

Diagnostics of energetic electrons with anisotropic distributions in solar flares

I. Hard X-rays bremsstrahlung emission

V. V. Zharkova[1], A. A. Kuznetsov[2], and T. V. Siversky[1]

[1] Department of Computing and Mathematics, University of Bradford, Bradford BD7 1DP, UK
e-mail: [v.v.zharkova;t.siverskyi]@brad.ac.uk
[2] Institute of Solar-Terrestrial Physics, Irkutsk, Lermontov str. 126a, 664033, Russia
e-mail: a_kuzn@iszf.irk.ru

From Electron Transport to the Fokker Planck Equation

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} + \left(\frac{\partial f}{\partial t}\right)_{\text{magn}},$$

The collisional integral is taken in the linearized form suggested by (Diakonov & Somov 1988)

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = \frac{1}{v^2}\frac{\partial}{\partial v}\left[v^2\nu(E)\left(\frac{k_B T_e}{m_e}\frac{\partial f}{\partial v} + vf\right)\right] + \nu(E)\frac{\partial}{\partial\cos\alpha}\left(\sin^2\alpha\frac{\partial f}{\partial\cos\alpha}\right),$$

where T_e is the electron temperature of the ambient plasma, k_B is the Boltzmann constant, and $\nu(E)$ is the frequency of collisions (Emslie 1978)

$$\nu(E) = \frac{k}{\sqrt{2m_e}}\pi n\lambda e^4 E^{-3/2},$$

where λ is the Coulomb logarithm for electrons, n is the ambient plasma density, and the parameter k defines beam electrons scattering on the ambient particles Emslie (1978).

Column Depth

$$\xi(z)=\int n(z)dz$$

From Electron Transport to the Fokker Planck Equation

$$\frac{\partial f}{\partial t} + v\cos\alpha\frac{\partial f}{\partial z} - e\mathcal{E}v\cos\alpha\frac{\partial f}{\partial E} - \frac{e\mathcal{E}\sin^2\alpha}{m_e v}\frac{\partial f}{\partial\cos\alpha} = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} + \left(\frac{\partial f}{\partial t}\right)_{\text{magn}},$$

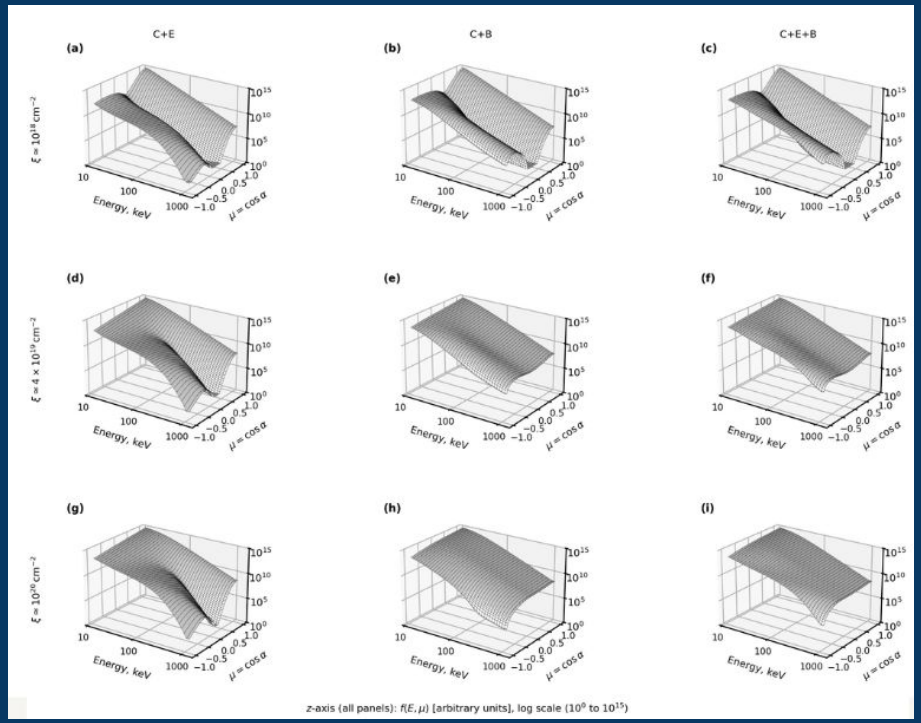
$$\left(\frac{\partial f}{\partial t}\right)_{\text{magn}} = \frac{1}{2}v\sin^2\alpha\frac{\partial \ln B(z)}{\partial z}\frac{\partial f}{\partial\cos\alpha},$$

where $B(z)$ is the magnetic field at a given depth z.

Column Depth

$\xi(z)=\int n(z)dz$

$$\frac{\partial \ln B}{\partial \xi} = \frac{1}{n}\frac{\partial \ln B}{\partial z} = \begin{cases} \frac{1}{\xi_c}\ln\frac{B_c}{B_0}, & \xi < \xi_c, \\ 0, & \xi \geq \xi_c, \end{cases}$$



Electron Distribution Function Vs. Column Depth for γ = 3.

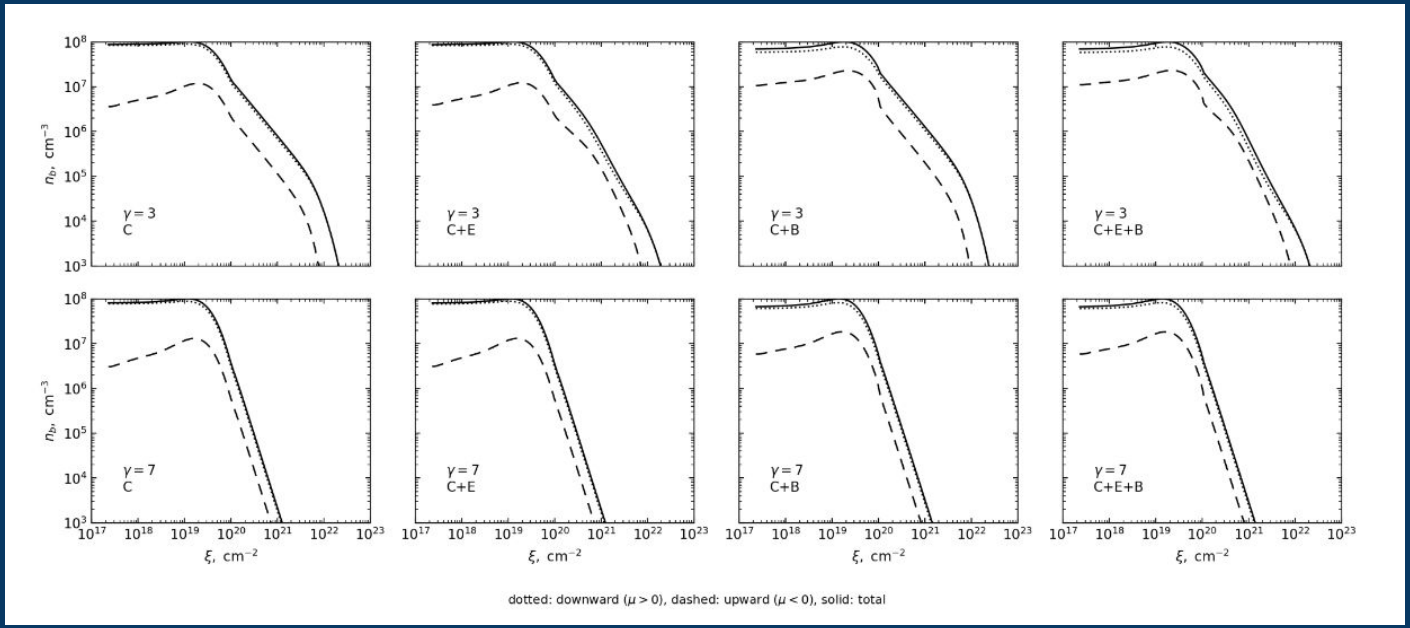
From the FP

- Near injection ($\xi \approx 10^{18}$): strongly forward-beamed ($\mu \rightarrow 1$).
- At $\xi \approx (4 \times 10^{19})$: pronounced backward lobe centered at $\mu \approx -0.7$: magnetic mirroring.
- Deeper ($\xi \approx 10^{20}$): distributions broaden further toward isotropy.

Beam Density vs.Column Depth

beam density (in cm^{-3}),

$$n_{\mathrm{b}}(\tau, s) = F_0 \sqrt{\frac{m_{\mathrm{e}}}{2E_0^3}} \int_{z_{\min}}^{z_{\max}} \mathrm{d}z \int_{-1}^{1} \mathrm{d}\mu \sqrt{z} A(s) f(\tau, s, z, \mu)$$



Now → Polarization

Monthly Notices of the ROYAL ASTRONOMICAL SOCIETY

MNRAS **461,** 2162–2173 (2016) doi:10.1093/mnras/stw1370
Advance Access publication 2016 June 9

Polarization of thermal bremsstrahlung emission due to electron pressure anisotropy

S. V. Komarov,[1,2]★ I. I. Khabibullin,[1,2] E. M. Churazov[1,2] and A. A. Schekochihin[3,4]

APPENDIX B: RELATIVISTIC BREMSSTRAHLUNG DIFFERENTIAL CROSS-SECTIONS

The expressions for the relativistic bremsstrahlung differential cross-sections were first given by Gluckstern & Hull (1953). After being rearranged in a more convenient way (Bai & Ramaty 1978) and corrected for typos, they read

$$d^2\sigma_\perp = A\left(B_\perp + \frac{L}{pp'}C_\perp + \frac{l_0}{p'Q}D_\perp - \frac{E}{p^2\sin^2\theta}\right), \tag{B1}$$

$$d^2\sigma_\parallel = A\left(\tilde{B}_\parallel(\theta) + \frac{L}{pp'}\tilde{C}_\parallel(\theta) + \frac{l_0}{p'Q}D_\parallel + \frac{E}{p^2\sin^2\theta}\right), \tag{B2}$$

$$A = \frac{Z^2}{8\pi}\frac{r_0^2}{137}\frac{p'}{p}\frac{1}{\epsilon}, \tag{B3}$$

$$B_\perp = -\frac{5\gamma^2+2\gamma\gamma'+1}{p^2\delta^2} - \frac{p^2-k^2}{Q^2\delta^2} - \frac{2k}{p^2\delta}, \tag{B4}$$

$$B_\parallel = -\frac{5\gamma^2+2\gamma\gamma'+5}{p^2\delta^2} - \frac{p^2-k^2}{Q^2\delta^2} + \frac{2(\gamma+\gamma')}{p^2\delta} - \frac{4l}{p'\delta}, \tag{B5}$$

$$\tilde{B}_\parallel(\theta) = B_\parallel + \frac{8(2\gamma^2+1)}{p^2\delta^4}\sin^2\theta, \tag{B6}$$

$$C_\perp = \frac{2\gamma^2(\gamma^2+\gamma'^2)-(5\gamma^2-2\gamma\gamma'+\gamma'^2)}{p^2\delta^2} + \frac{k(\gamma^2+\gamma\gamma'-2)}{p^2\delta}, \tag{B7}$$

$$C_\parallel = \frac{2\gamma^2(\gamma^2+\gamma'^2)-(9\gamma^2-4\gamma\gamma'+\gamma'^2)+2}{p^2\delta^2} + \frac{k(\gamma^2+\gamma\gamma')}{p^2\delta}, \tag{B8}$$

$$\tilde{C}_\parallel(\theta) = C_\parallel + \frac{4\gamma(3k-p^2\gamma')}{p^2\delta^4}\sin^2\theta, \tag{B9}$$

$$D_\perp = \frac{k}{\delta} - \frac{k(p^2-k^2)}{Q^2\delta} + 4, \tag{B10}$$

$$D_\parallel = \frac{4}{\delta^2} - \frac{7k}{\delta} - \frac{k(p^2-k^2)}{Q^2\delta} - 4, \tag{B11}$$

$$E = \frac{2L}{pp'}\left(2\gamma^2-\gamma\gamma'-1-\frac{k}{\delta}\right) - \frac{4l_0}{p'Q}(\delta-\gamma')^2 - \frac{2l(\delta-\gamma')}{p'}, \tag{B12}$$

where

$$\gamma = E/m_ec^2+1; \quad \gamma' = \gamma - \epsilon/m_ec^2; \tag{B13}$$

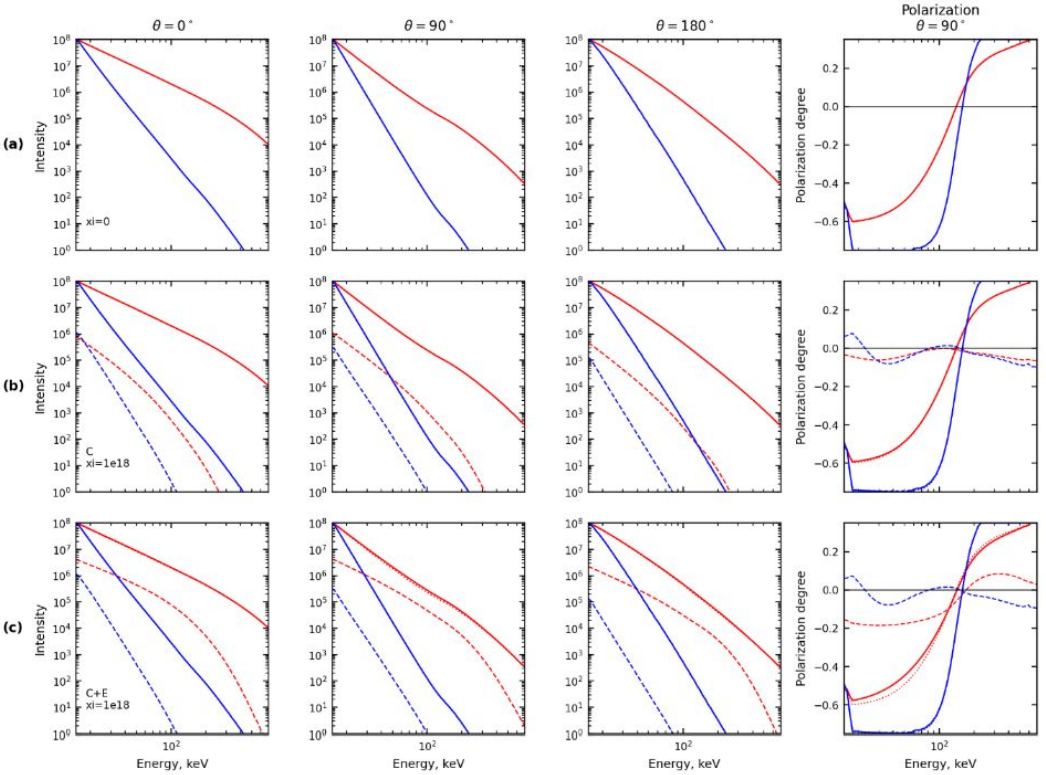
$$p = \sqrt{\gamma^2-1}; \quad p' = \sqrt{\gamma'^2-1}; \quad k = \epsilon/m_ec^2; \tag{B14}$$

$$Q^2 = p^2+k^2-2pk\cos\theta; \quad \delta = \gamma - p\cos\theta; \tag{B15}$$

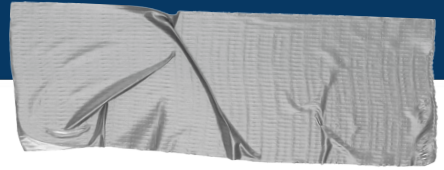
$$L = 2\ln\left(\frac{\gamma\gamma'+pp'-1}{\gamma\gamma'-pp'-1}\right); \quad l = \ln\left(\frac{\gamma'+p'}{\gamma'-p'}\right); \quad l_0 = \ln\left(\frac{Q+p'}{Q-p'}\right); \tag{B16}$$

$r_0 = e^2/(m_ec^2) \approx 2.82\times10^{-13}$ cm is the classical electron radius, the rest of the notations were introduced in Section 2.2.

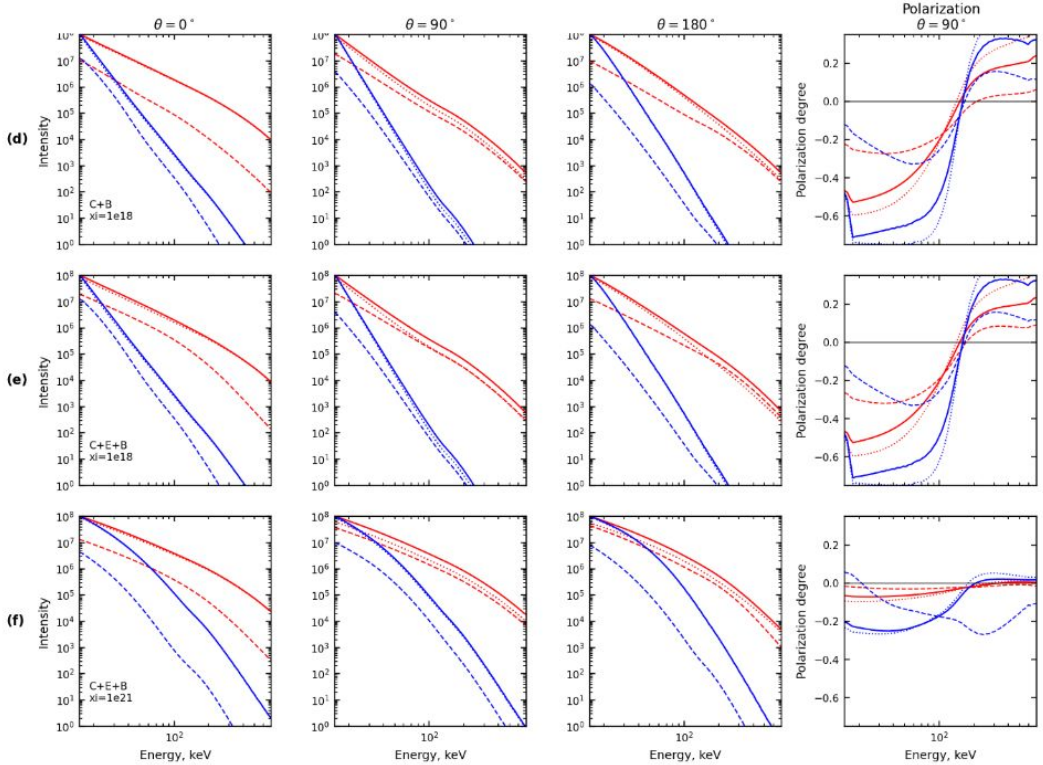
This paper has been typeset from a TEX/LATEX file prepared by the author.



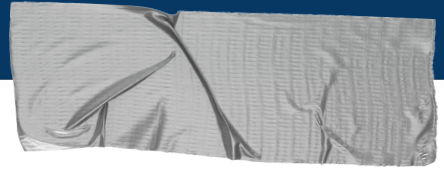
Hard X-ray Spectra and Polarization Degree Across Depth



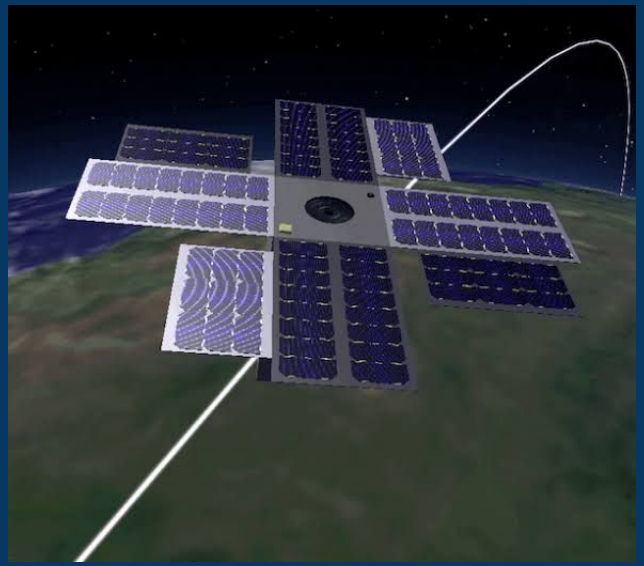
- Top row: injected spectrum before atmospheric transport
- Columns 1–3: emitted intensity at three viewing angles (θ=0°,90°,180°); red/blue compare γ=3 vs. γ=7.
- Column 4: degree of polarization at θ=90° vs. photon energy.
- Polarization weakens with depth.
- Electron directivity leaves a direct, measurable imprint on HXR polarization.

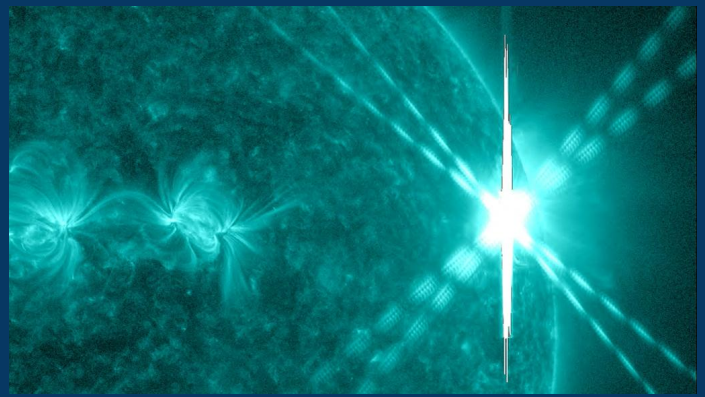


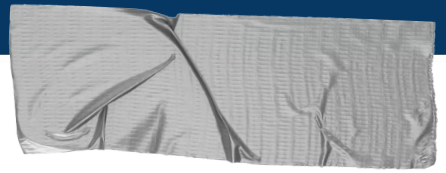
Hard X-ray Spectra and Polarization Degree Across Depth



- Top row: injected spectrum before atmospheric transport
- Columns 1–3: emitted intensity at three viewing angles (θ=0°,90°,180°); red/blue compare γ=3 vs. γ=7.
- Column 4: degree of polarization at θ=90° vs. photon energy.
- Polarization weakens with depth.
- Electron directivity leaves a direct, measurable imprint on HXR polarization.







Further Work

- Physical model refinements (more realistic magnetic field profile).
- Explore different injection regimes.
- Polarization: compare with observational data (PADRE - Sharp).

Thanks!